BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 286 · Lossy and lossless compression

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What can a firewall block?

Q2. What is non-volatile storage?

Q3. What is a bitwise mask?

Q4. Apply bitwise AND: $11011010_2 \wedge 00001111_2$. Give 8-bit answer and hexadecimal.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

lossless — Reconstructs original data exactly.

lossy — Permanently removes some information.

compression ratio — Original size compared with compressed size.

redundancy — Repeated/predictable information.

decompression — Recover encoded content.

New learning

Understand — Lossless compression represents data more efficiently without discarding any information and reconstructs the original exactly.

Apply — Lossy methods intentionally discard information; a decoded image/audio can be perceptually similar but not bit-for-bit identical.

Evaluate — Choose lossless for text, programs, records or medical numeric originals; lossy can be appropriate for delivery copies of photos/video/audio with acceptable quality reduction.

Worked example

A database contains precise 7-digit identifiers. Is lossy compression suitable?

1. Predict first — An altered digit can identify a different record and corrupt its meaning.

2. Model and check — Lossless encoding is essential; compression ratio depends on redundancy in the source.

We Do · Practise with guidance

A web page uses 4K source photographs but displays small thumbnails. Evaluate lossy compression.

```
Original archive = lossless/source copy
Web thumbnail = lossy resized/encoded version
```

Predict first

A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.

Model and check

Keep originals separately so discarded detail can be recovered for future editing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define lossy and lossless compression; state one suitable use of each. [6]

Question 2

Given sequence A A A B B A, write RLE run/value pairs and show how to decode. [6]

Question 3

A school distributes music and exam marks electronically. Recommend encoding and justify trade-offs. [6]

Exit Ticket · Answer before checking

Lossy compression is suitable for any data if it produces the smallest file.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Traffic failing its security rules.

Q2. Storage retaining content without power.

Q3. Bit pattern applied with bitwise operators to select/change bits.

Q4. $00001010_2 = 0A_{16}$ (last four bits 1010 retained).

You Do pages · Independent questions

Q1. Lossless exact reconstruction [2], use source/text/records [1]; lossy discards information [2], use delivery media [1].

Q2. (A,3),(B,2),(A,1) [3]; three A [1], two B [1], one A [1].

Q3. Audio lossy if acceptable sound/size [2]; exam marks lossless [2]; lossy removes exact detail [1], preservation of originals/recovery [1].

Exit Ticket · Model answer

False. Some data must remain exact.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: